TSM 099 Quadratic and Rational Equations Test - Practice 1

Version 202610122020.25 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Solve. List excluded values and check in the original equation.

$$\frac{4}{2x-2} + \frac{7}{x^2-5x+4} = \frac{3}{x-4}$$

2. Solve. List excluded values and check candidates in the original equation.

$$\frac{4}{x^2-4} + \frac{x-3}{x-2} = \frac{5}{x+2}$$

3. Solve by factoring.

$$x^2 + 3x - 4 = 0$$

4. Solve by taking square roots; do not expand the square.

$$2(x+1)^2 = 8$$

5. Solve with the quadratic formula. Simplify radicals.

$$2x^2 + x - 2 = 0$$

Student answer sheet

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Question 1

Excluded values	5 pts
Solution/check	15 pts

Question 2

Excluded values	4 pts
Candidates/check	6 pts
Solution	10 pts

Question 3

Solutions	20 pts

Question 4

Solutions	20 pts

Question 5

Solutions	20 pts

Answer key

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Question 1

Excluded values	$\{1, 4\}$	5 pts
Solution/check	$x = 2$	15 pts

Question 2

Excluded values	$\{-2, 2\}$	4 pts
Candidates/check	$x = 2$ (rejected), $x = 4$	6 pts
Solution	$x = 4$	10 pts

Question 3

Solutions	$x = -4,\ 1$	20 pts

Question 4

Solutions	$x = -3,\ 1$	20 pts

Question 5

Solutions	$x = -\frac{1}{4} + \frac{\sqrt{17}}{4},\ \ -\frac{\sqrt{17}}{4} - \frac{1}{4}$	20 pts